MING-FENG HO (何銘峰)

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CURRENT POSITION

· Postdoctoral Fellow in the Leinweber Center for Theoretical Physics, University of Michigan *2024 Sept -*
Mentor: Camille Avestruz

EDUCATION

· Ph.D. student in Physics & Astronomy, University of California, Riverside *2018 - 2024 Aug*
Research advisor: Prof. Simeon Bird
Thesis: Model-driven cosmology with Bayesian machine learning and population inference
· M.S. in Astrophysics, National Taiwan University *2016 - 2018*
· B.S. in Physics, National Taiwan University *2010 - 2014*

RESEARCH INTERESTS

· **Cosmology**: Cosmological inference, hydrodynamical simulations, IGM as a cosmological probe.
· **Bayesian statistics**: Gaussian process, multi-fidelity, hierarchical inference, Bayesian deep learning.
· **IGM/CGM**: Damped Lyα absorbers (DLAs), metal absorbers, quasar emission modeling, redshift inference.
· **Black holes**: Population inference for gravitational waves, primordial black holes (PBHs).

AWARDS AND GRANTS

1. NASA FINESST Fellowship *2021 - 2024*
2. Anne Kernan Award (Outstanding Senior Graduate Student Researcher), UC Riverside *2023*
3. Provost's Scholars Fellowship (honored), UC Riverside *2021*
4. Benjamin C. Shen Award (Outstanding Junior Graduate Student Researcher), UC Riverside *2021*
5. GSA Conference Travel Award, UC Riverside *2019*
6. Student Thesis Award, Physics Society of Taiwan *2019*
7. Dean's Fellowship, UC Riverside *2018*
8. Laureate for Philosophical Treatise, National Taiwan University *2012*

RESEARCH MENTORSHIP

Advisor for UCR/UCLA undergraduates advised by Prof. Simeon Bird

· Kevin Hong (co-advised with Sum Qezlou): Animating the evolution of universe using `Blender` [YouTube Video] *2022*
· Ryan Tsaio: Automated detection of Lyman limit systems *2023*

Research advisor for King's high school students

· Emma Shah (now UC Berkeley; co-advised with Phoebe R. Upton Sanderbeck): DLA spatial separation on the detection pipeline *2020 - 2021*
· Emma Shah and Rafael Rosales: A better prior for QSO redshift estimation (GitHub `jibanCat/gpy_dla_detection`, Gold medal in the county-level science fair) *2021 - 2022*

Team Lead for Data Science Challenge, Lawrence Livermore National Laboratory

· Adhith Karthikeyan, Alex Chen, Matthew Lee: Deep learning for galaxy/asteroid *2021*

PUBLICATION

Summary

· 15 publications, 15 refereed (114 total citations; Google scholar 09/17/2024).
· 5 first-author publications (80 total citations).
· 3 second-author publications with major contributions (involving writing and software development).
· 3 third-author publications with important software contributions.
· 4 collaboration papers.

SCIENTIFIC PRESENTATIONS

Summary

· 20 talks, posters and presentations since 2019
· 6 remote talks, 8 in-person seminar talks, 1 poster, 5 conference talks

Selected Talks

· *Invited Talk: CfA Seminar (topic: Multi-fidelity emulators and cosmological constraints)* *Nov 2023*
· *Invited Talk: IPMU APEC seminar (topic: PRIYA emulator using Astrid simulations)* *Oct 2023*
· *Contributed Talk: Royal Astronomical Society 1D ML (topic: GP-DLA finder)* [YouTube Video] *Mar 2023*
· *Contributed Tutorial: KITP Galaxy ML Workshop (topic: GP emulator)* *Feb 2023*
· *Invited Talk: KICP seminar (topic: MF emulator)* *Jan 2023*
· *Contributed Talk: CCA's CAMELS Workshop (topic: MF emulator using Astrid)* *Nov 2022*

SERVICE

Selected service

· Referee for Physical Review D, Astrophysical Journal, and MNRAS *2021 - present*
· P&A PeER Mentorship (PAPER) Leader *2023 - present*
· P&A Student Seminar (PASS) Founder *2022 - 2023*
· Head Steward (Riverside), UAW 2865 *2021 - 2022*
· UC Astronomy Osterbrock Sierra Conference Co-I *2021*

PUBLIC OUTREACH

1. UCR's Stargazing Series Presenter and Moderator (× many times) [YouTube Video] *2020 - 2021*
2. UCR's Mercury Transit Presenter (hands-on demos) *2019*

WORKING EXPERIENCE

Research Assistant for Digital Humanities Taipei, Taiwan

· Text mining for digital humanities, working at Academia Sinica, Chinese literature and Philosophy *2018*

Magazine Editor/Freelance Writer Taipei, Taiwan

· Editing for Little Newton, and also writing literature, receiving the nation-wide Hakka literature award *2015*

Taiwanese Military Service Keelung, Taiwan

· Digitalize historical court records *2014 - 2015*

OTHER SKILLS

· Languages: Mandarin (native), English, Japanese (limited listening/reading), Taiwanese (limited listening)